ITCC · FTCC · NOVEMBER 2026

Keep your tools. See their history.

GitHub for IT Work

[Speaker name], [title]

tinyurl.com/itccgh26

norrisaftcc.github.io/itcc26

- Add printer fix steps
- Update VLAN notes
- Log IOCs from report
- Add build checklist

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THE WRONG-VERSION PROBLEM

One folder, five résumés

```
resume.docx
resume_final.docx
resume_final_v2.docx
resume_final_v2_REAL.docx
resume_USE_THIS_ONE.docx
```

Which one did you send?

Have you heard “that’s the wrong version”?
I can show you a way to manage that.

EVERY TRACK HAS THIS FOLDER

The problem is not only résumés

CYBER

`ioc-notes-FINAL.txt`

Indicators logged from phishing reports

NETWORK

`switch-config-OLD-works.txt`

The lab switch config that last worked

HELP DESK

`printer-fix-steps-COPY(2).docx`

Steps to clear a stuck print queue

HARDWARE

`pc-build-checklist-v3.docx`

Parts and POST checks for a lab build

TODAY, IN THREE ACTS

Then and now. Show me. Your turn.

ACT 1

Then and now

Three eras of keeping code

What we were missing

What GitHub is

about 7 min

ACT 2

Show me

One résumé, two repos

One phish, four looks

One deleted password

about 25 min

ACT 3

Your turn

Your first commits, in pairs

Better or worse?

One thing for this week

about 25 min

1996 TO TODAY

Three eras of keeping code

Where does the file live? Who can change it? How do you undo a mistake?

1996

A shared drive

Lives

A read-write Windows NT share

Who

Anyone

Undo

“Get the tapes.”

→

Late 1990s to 2012

CVS · SVN · ClearCase

Lives

A server, one main line

Who

Everyone, on main

Undo

Careful hands. “Don’t break the build.”

→

Today

Git + GitHub

Lives

Your laptop and GitHub

Who

Anyone proposes. The owner decides.

Undo

History and revert

WHAT WE WERE MISSING

We had the rule. We did not have the machine.

Martin Fowler, “Continuous Integration”

First published 10 September 2000

1. Everyone pushes commits to the mainline every day.
2. Every push to mainline should trigger a build.
3. Fix broken builds immediately.

> *“nobody has a higher priority task than fixing the build.”*
>
> Kent Beck, quoted by Fowler

2009: the word “DevOps” arrives. Today it is a whole field.

Source: martinfowler.com/articles/continuousIntegration.html (2000; rewritten 2006 and 2024). DevOps.com, “The Origins of DevOps.”

WHAT GITHUB IS

A burger. Also a chocolate cake.

6	**AI**	Copilot	
5	**Security**	code scanning, push protection	one bite
4	**Automation**	Actions, Pages, Codespaces	
3	**Working together**	pull requests, issues, Projects	
2	**Hosting**	your repo and its history, online	
1	**Git**	a tool on your computer	

Today we eat the burger.

Layers 1 to 3, plus one bite of layer 5.

SIX WORDS

Six words cover today

repo

A project folder on GitHub

commit

A saved change, with a note

history

The list of every commit

diff

The changed lines: + added, – removed

pull request

A proposed change. The owner decides.

push

Send commits from your computer to GitHub

DOES GITHUB GET YOU HIRED?

I found no evidence that help-desk employers check GitHub.

47%

named certifications a critical quality (IT experience: 44%)

ISC2 hiring study, 2025. 929 cyber hiring managers, 6 countries, surveyed Dec 2024.

43%

named documentation, such as processes, a top entry-level task

Same ISC2 study, as reported by Infosecurity Magazine, June 2025.

Use GitHub because it keeps your tools organized and shareable.

If an employer looks, that's a bonus.

Two questions, not a ranking. Neither question is about GitHub. Both are about cyber hiring only.

DEMO 1 · 6 MINUTES

One résumé, two repos

jordan-resume PRIVATE

Full résumé: phone, email, every version.

Save as PDF, then send the file.

itcc26/jordan

Name, city, certifications, links. Fictional student.

No phone. No street address. Anyone can see it.

```
resume.md · Add printer mapping to help-desk duties
+ - Map network printers for two campus computer labs
1 line added · 0 removed · the newest commit is the one you send
```

ACT 2 · A SECURITY CASE

One phish, four looks

1. **Spot it**
   Three red flags you can see in any reported email.
2. **Triage it**
   A script, a runbook, and a prompt, kept in one repo.
3. **Fix it**
   A web page shows the sender name, with a bug. The fix is one commit.
4. **Same cause, older code**
   The same mistake in C: a buffer overflow.

We show each bug and its fix. No attacks today.

THREE RED FLAGS

Look for three things only

From and Reply-To differ

Replies go somewhere else

```
From: security@paypa1-secure.example.net
Reply-To: verify-desk@example.org
```

Link text and link differ

What you see is not where it goes

```
Shows: www.example.com
Goes to: login.paypa1-secure.example.net
```

SPF fail

The mail server checks this for you

```
SPF fail
```

The sender is not allowed to send for the sending domain.

DEMO 2, PART 1 · 2.5 MINUTES

The phishing-triage kit

```
Codespace terminal · real output, middle lines omitted (…)
$ python3 scripts/parse_headers.py samples/fake-phish.eml
…
Red flags:
  3  [!] SPF result is 'fail'.
skip [!] DMARC result is 'fail'.
  1  [!] Reply-To domain (example.org) differs from From domain (paypa1-secure.example.net).
  2  [!] Link text shows 'www.example.com' but the link goes to 'login.paypa1-secure.example.net'.
```

Next: the same sender name, shown on a web page. That page has a bug.

DEMO 2, PART 2 · 5 MINUTES

The fix is a commit

Phish Report Viewer (lab)

Sender display name

`<b>Security Team</b>`

Before fix **Security Team**

After fix `<b>Security Team</b>`

```
Pull request · fix-sender-text · index.html
- output.innerHTML = nameInput.value;
+ output.textContent = nameInput.value;
```

innerHTML reads the name as HTML. textContent reads it as plain text.

1 line removed · 1 line added

Bug
Name runs as HTML

→

Alert
If CodeQL flags it

→

Pull request
A one-line fix

→

Merge
Tags show as text

1 MINUTE

Same cause in C

vulnerable.c → fixed.c

```
char name[16];
- strcpy(name, argv[1]);
+ snprintf(name, sizeof name, "%s", argv[1]);
```

1 line removed · 1 line added



The program writes past the end of a 16-byte buffer. Security+ calls this a buffer overflow.

AddressSanitizer: stack-buffer-overflow · WRITE of size 41 — the fixed copy prints “Hello, AAAAAAAAAAAAAAA” and stops at the edge.

DEMO 3 · 4 MINUTES

Is the password gone?

commit 1 · router-backup.txt

Add lab router backup

→

commit 2 · file deleted

Remove router backup

→

vote

Hands up: is it gone?

History · Remove router backup · still readable

```
- enable secret 5 $1$mK4q$PcTFsirRkEscGyZPfWSxY0
- snmp-server community LabR0Community RO
- password 7 0815494F0A1155191E124D272525
```

A hash or a type 7 string is not safe to publish. Type 7 is reversible.

GIT REMEMBERS EVERYTHING

The history that saves your résumé keeps the password

~29M

new hardcoded secrets in public GitHub commits in 2025

64%



of secrets confirmed valid in 2022 were still valid four years later

Whose data is it? Do not put campus or employer configs in a personal repo without permission.

Source: GitGuardian, State of Secrets Sprawl 2026. GitGuardian sells secret scanning.

PUSH PROTECTION

Catches some, misses some

Catches

Structured provider tokens, such as AWS keys

Misses

Plain passwords, such as an enable secret, on free personal repos

Source: GitHub Docs, “Supported secret scanning patterns”: passwords are AI-detected patterns. The capability table lists no push protection for them. Checked Nov 2, 2026.

AI USE

Label it. Guard the key.

talk-itcc-phishing-triage / README.md

AI use

Tool: Claude (Anthropic).

Produced: First drafts of the script, runbook, prompt, and samples.

Verified: Script tested on two sample emails; output in TEST-OUTPUT.md.

Gmail menu steps in runbook.md checked against Google Help on 2026-10-26.

Outlook on the web menu names not checked; the runbook gives none.

Reviewed by: [presenter, date]



An AI tool's API key is a secret, like a password.

Keep it out of every commit.

HANDS-ON · 18 MINUTES + 4 BUFFER · IN PAIRS

Your first commits

1. Open **talk-itcc-starter** from the talk page. Use this template. Private.
2. Pick one track file.
3. Click the pencil. Edit one line.
4. Commit, with a message that says what changed.
5. Open the track file, then History: 2 commits.
6. In README.md, fill the AI-use block truthfully, then the practice block.
7. Your partner finds the commit with your message.
8. Swap seats. The reader edits a second file.

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1	2	3	4	5	6	7	8 swap	buffer

THIS WEEK

One real tool, one repo

Put one tool you already use in a repo.

Write its README, with the AI-use lines.

Never commit a secret.

TAKE HOME · LINKS ON THE TALK PAGE

GitHub Skills “Introduction to GitHub” · a Projects board for your job hunt · your own copy of talk-itcc-phishing-triage · the talk-itcc-xss-fix-lab exercise

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